

LITERATURE REVIEW ON AUTONOMOUS UNDERWATER VEHICLE (AUV) SIMULATION USING GAZEBO

**A Project Report Submitted to the Defence Research and
Development Organisation (DRDO)**



**Institute for Systems Studies & Analyses (ISSA)
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Information Technology

ABOUT THE ORGANIZATION

Defence Research & Development Organisation (DRDO)

The Defence Research & Development Organisation (DRDO) operates under the Department of Defence Research and Development, part of the Ministry of Defence. DRDO is committed to enhancing self-reliance in defense systems by designing and developing world-class weapon systems and equipment. These advancements align with the expressed needs and qualitative requirements of the three services: The Army, Navy, and Air Force.

DRDO was established in 1958 through the merger of:

The Technical Development Establishment (**TDES**) of the Indian Army, The Directorate of Technical Development and Production (**DTOP**), and The Defence Science Organisation (**DSO**).

The Defence Research and Development Organisation (DRDO) is a network of more than 50 laboratories, dedicated to advancing defense technologies across various domains. These include aeronautics, armaments, electronics, combat vehicles, engineering systems, instrumentation, missiles, advanced computing and simulation, special materials, naval systems, life sciences, training, information systems, and agriculture. The organization is involved in several major projects, focusing on the development of missiles, armaments, light combat aircraft, radars, electronic warfare systems, and more. DRDO has already achieved significant milestones in many of these technologies. As an agency of the Republic of India, DRDO is responsible for the military's research and development, with its headquarters located in New Delhi. In addition to meeting the military's cutting-edge technology requirements, DRDO's innovations provide considerable spin-off benefits to society at large, contributing to national development and strengthening India's defense capabilities.

Vision

To make India prosperous by establishing a world-class science and technology base, and to provide our Defense Services with a decisive edge by equipping them with internationally competitive systems and solutions.

Mission

- To design, develop, and lead the production of state-of-the-art sensors, weapon systems, platforms, and allied equipment for our Defense Services.
- To provide technological solutions to the Services, optimizing combat effectiveness and promoting the well-being of the troops.
- To develop infrastructure, nurture committed quality manpower, and build a strong indigenous technology base.

Institute for Systems Studies & Analyses (ISSA)

ISSA is involved in System Analysis, Modeling & Simulation for various defence applications pertaining to employment/deployment, tactics & force potential evaluation, tactical/strategic & mission planning etc. We develop war games for all the three Services.

Consequent to the reorganization of the system analysis activities within DRDO, the functions of DSE were redefined and was given the present name, i.e. Institute for Systems Studies & Analyses (ISSA) in the year 1980. In the year 1972, a small group named the Aeronautical Systems Analysis Group (ASAG) was created with an objective to carry out aeronautical systems studies and analyses.

This group was functioning from National Aeronautics Laboratory, Bangalore as a detachment of the Directorate of Aeronautics. In the year 1974, ASAG was converted into a self-accounting full-fledged unit named as Centre for Aeronautical Systems Studies & Analyses (CASSA) and was shifted to the premises of Aeronautical Development Establishment, Bangalore.

From 1974 to 2003, CASSA contributed significantly to a series of systems analysis studies attributed to DRDO HQrs and the Indian Air Force. It contributed into several design and development activities and policy level issues with its objective analyses.

In the year 2003, CASSA was merged with ISSA with an objective to synergies systems analysis activities and wargaming development processes under integrated combat environment. With this, ISSA has emerged as a nodal systems analyses lab and in 2013 ISSA is placed under SAM Cluster with the mandate in the field of Training & Planning Wargames, Integrated Air Defence & EW, Combat Modeling, Simulation System Analysis and computerised Wargame.

Area Of Work

- Modeling, Simulation, Systems Analysis for Defence Application
- Systems Evaluation Studies
- Combat Model Development
- Computer Science and Applications for Defence Modelling & Simulation Domain
- Operations Research and Decision Support Techniques.

Vision: Transform ISSA into a centre of excellence in system analysis, modelling & simulation of defence systems to meet the challenges of the present and future requirements of the armed forces.

Mission: Conduct system study and develop high quality integrated software for system analysis & decision support in application areas of Sensors & Weapons, Electronic Combat, Land & Naval Combat, Air-to-Air Combat and Air Defence for effective use by DRDO and Services for Design, Mission Planning, Tactics development and Training.

DECLARATION

I hereby declare that this submission is my own work. While it includes a literature survey based on previously published works, I have appropriately acknowledged all sources within the text and references. To the best of my knowledge and belief, this work contains no material authored by another individual that has not been properly cited. Furthermore, it does not include any content that has been substantially accepted for the award of any degree or diploma from any university or other institution of higher learning, except where due acknowledgment has been appropriately made.

Signature

Nishita Naman

CERTIFICATE

This is to certify that the Internship/Training Report titled " **LITERATURE REVIEW ON AUTONOMOUS UNDERWATER VEHICLE (AUV) SIMULATION USING GAZEBO**", conducted from 1/06/2026 to 31/07/2026 submitted by **Nishita Naman** as part of the requirements for the B.Tech degree in the Department of **Information Technology** at **Indira Gandhi Delhi TEchnical University for Women** is an original record of the candidate's work carried out under my supervision.

I hereby declare that, to the best of my knowledge and belief, this submission represents the candidate's own work. Furthermore, it does not include any content that has been substantially accepted for any other degree or diploma from a university or other institute of higher learning, except where appropriate acknowledgments are provided in the text.

Neha Aggarwa;

ISSA, DRDO

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1.Introduction

1.1 Autonomous Underwater Vehicle

Autonomous Underwater Vehicle (AUV) is a robotic ,uncrewed submarine that drives itself through water to complete tasks without requiring real time control from a human ,AUV is completely untethered ie not bound to something ,as it relies on its own onboard computers ,power and sensors to navigate and survive the extreme conditions of the deep ocean

Unlike **Remotely Operated Vehicles (ROVs)** , which are physically connected to a surface ship by a cable and driven by a human with a joystick, an AUV is structurally and operationally independent . They are self powered using advanced lithium ion battery matrices and control their behaviour through onboarding computational architectures and pre-programmed flight algorithms.

Since regular GPS and Wi-Fi signals die out within just a few meters underwater, Autonomous Underwater Vehicles (AUVs) can't navigate the way we do on land. Instead, they figure out exactly where they are in 3D space by combining data from acoustic sensors and internal motion trackers, primarily Inertial Measurement Units (IMUs) and Doppler Velocity Logs (DVLs). Eliminating the physical tether provides a significant operational advantage for marine operations : it cuts out cable drag, frees the robot from needing a massive support ship hovering right above it, and allows it to safely explore deep or dangerous areas where a cable could easily get tangled.

1.2 Applications of AUVs

Because they can operate on their own, travel long distances, and carry a wide variety of sensors, AUVs have become essential tools for military, scientific, and commercial work. Their primary applications include:

- **Defense and Strategic Operations:** AUVs are crucial for finding underwater mines (Mine Countermeasures, or MCM). Navies use them to safely map and identify explosive threats using advanced sonar, meaning human divers and surface ships don't have to be put in danger. They are also used to continuously gather intelligence, automatically track targets, and quickly assess the environment in disputed coastal areas.
- **Scientific and Oceanographic Research:** In the academic world, AUVs are the go-to tool for exploring the seafloor. Researchers use them to create highly detailed underwater maps, survey hydrothermal vents with cameras and sound, and collect data on water chemistry. Because they aren't tied to a surface cable, they can even run missions deep under polar ice shelves—areas that regular research ships simply cannot reach.
- **Commercial and Industrial Infrastructure:** The offshore energy and telecom industries rely heavily on AUVs to inspect underwater structures. This includes subsea oil and gas pipelines, massive internet cables crossing the ocean, and the underwater foundations of offshore wind farms. Using cameras and sonar, an AUV can quickly scan miles of infrastructure to spot damage, erosion, or leaks. This is done much faster at a lower cost than using traditional, cable-controlled robots (ROVs).

1.3 Need for Simulation Before Sea Trials

Moving a robot from theoretical computer code to the actual ocean comes with major logistical, financial, and technical challenges. Unlike testing robots on land, the underwater environment is completely unpredictable and constantly shifting.

Running physical sea trials is incredibly expensive. It requires surface support ships, specialized equipment to launch and recover the robot (LARS), and a highly trained crew. On top of that, training an AUV's software to avoid obstacles, map its surroundings (SLAM), or learn how to navigate often requires thousands of trial runs.

This is where simulation becomes a vital stepping stone. By testing the robot's software in a virtual environment first, developers save huge amounts of time and money, drastically reducing the need for expensive hardware and constant, risky field tests.

Factor	Real world sea trials	Gazebo simulation
Data Acquisition	Slow ,bounded by battery life and logistics	Instantaneous ,scriptable and scannable
Edge Case Testing	Dangerous ;risking the loss of a highly expensive robot	Safe ;collision simply reset the environment
Ground Truth	Difficult to acquire accurately without external acoustic beacons	Absolute state parameters (x,y,z velocity) are always known for model validation.

1.4 Why Gazebo is Used for AUV Simulation

Gazebo has become the industry standard for underwater robotics because it offers a flexible, physics-based virtual world that connects directly with the Robot Operating System (ROS). For machine learning and control systems engineers, Gazebo provides three essential features:

- **Highly Realistic Physics:** It calculates complex 3D movement (6-DOF) to accurately recreate real-world physical forces. This means it can simulate exactly how water resistance, buoyancy, and ocean currents will push and pull on the robot.
- **Realistic Sensor Noise:** Real-world marine environments are subject to severe acoustic and optical disturbances . Gazebo lets developers simulate fading light, murky water, and sound interference. This is crucial for adding realistic "noise" to training data, ensuring that computer vision and sensor algorithms learn to handle imperfect, real-world conditions instead of just getting used to "perfect" simulated data.
- **Seamless Code Deployment:** Because Gazebo is fully integrated with ROS, developers don't have to change their code when moving from simulation to the real world. A Python script for machine learning or a C++ program for mapping (SLAM) will interact with the virtual robot the exact same way it would with the actual, physical hardware.

Section 2: Overview of the Gazebo Simulator

2.1 Gazebo Architecture

Gazebo operates on a decoupled (separated) client-server architecture .It is divided cleanly into a computational backend (`gzserver`) and a graphical frontend (`gzclient`). The `gzserver` handles the core execution loop, including physics computation, sensor generation, and state updates like the brain of the simulator .Whereas `gzclient` handles the 3D rendering and user interface via the Object-Oriented Graphics Rendering Engine (Ogre). Because these two parts are split, you don't always have to run them together. It allows researchers to run computationally intensive simulations, such as reinforcement learning training loops or

multi-robot swarm missions, in a "headless" mode without rendering graphics, thereby saving significant CPU and GPU overhead.

To define marine environments and physical systems, Gazebo employs the Simulation Description Format (SDF). Unlike the Unified Robot Description Format (URDF), SDF supports nested models, advanced friction properties, and custom plugins, making it suitable for the complex rigid-body structures and hydrodynamic profiles characteristic of underwater vehicles.

2.2 Physics Engines

Gazebo provides modular support for multiple physics engines, including the Open Dynamics Engine (ODE), Bullet, DART, and Simbody. By default, ODE manages rigid-body dynamics and contact collisions. However, standard physics engines do not natively calculate fluid-structure interactions specific to underwater environments. To address this, marine robotics researchers employ specialized simulation packages:

- **The `uuv_simulator` Package:** Developed by Manhães et al. (2016), this package utilizes custom Gazebo plugins to calculate hydrodynamic forces based on Fossen's equations. It treats fluid forces as external disturbances applied to rigid-body links.
- **The `freefloating_gazebo` Package:** In contrast to the link-by-link plugin forces of `uuv_simulator`, the `freefloating_gazebo` package alters fluid dynamics modeling at the foundational engine level. It computes buoyancy and fluid resistance directly from the geometry and submerged volume of the vehicle's links, making it highly effective for simulating underwater structures that cross the air-water interface or exhibit irregular surface geometries.

2.3 Sensor Simulation Capabilities

Simulating an AUV requires accurate modeling of aquatic sensor payloads, which are susceptible to signal degradation due to acoustic and visual disturbances (Lakhyari et al., 2024). Through frameworks such as the Aquatic Virtual Environment (DAVE) project, Gazebo replicates these specialized marine instruments (Zhang et al., 2022):

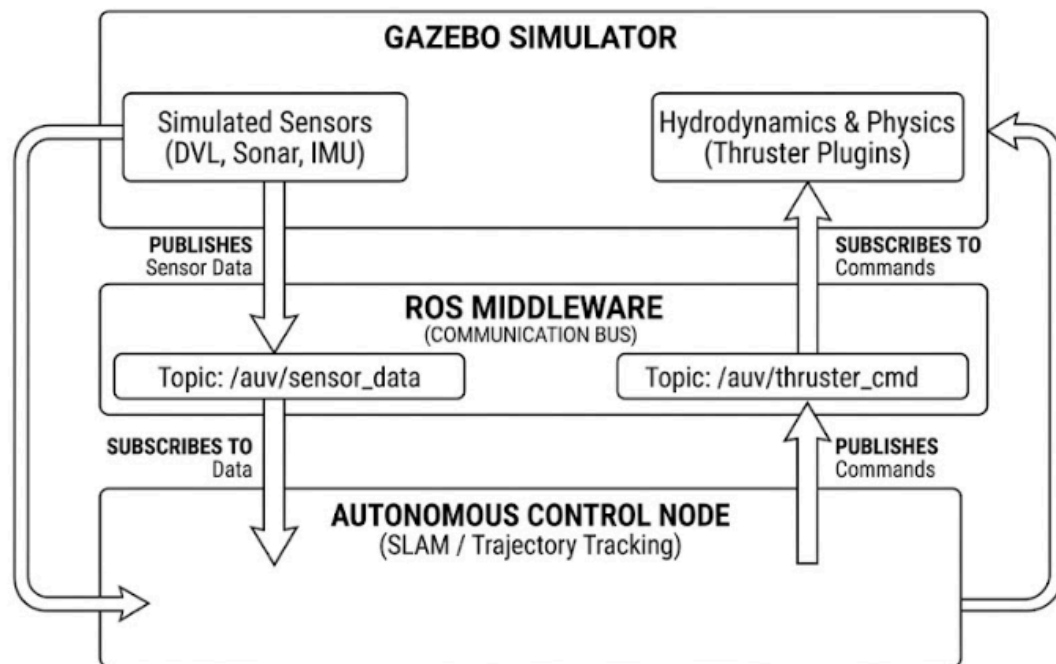
- **Doppler Velocity Logs (DVLs):** Simulated by casting ray vectors from the AUV model to the seafloor, measuring Doppler frequency shifts to compute ground-relative 3D velocity vectors.
- **Acoustic Sonars:** Packages like DAVE utilize GPU-accelerated ray-tracing to simulate multibeam, forward-looking, and side-scan sonars, generating realistic raw acoustic backscatter data and point clouds.
- **Inertial Measurement Units (IMUs):** Simulates high-frequency angular velocities and linear accelerations, complete with configurable bias drift and Gaussian noise models to replicate real-world physical errors.

Gazebo supports structural noise injection directly within its sensor plugins. For visual sensors, developers can configure shaders to simulate water turbidity, light absorption, and scattering effects. For acoustic instruments, high-frequency stochastic noise models are injected into the data streams, ensuring that downstream computer vision and sensor fusion algorithms are resilient to the imperfections of real-world underwater environments.

2.4 ROS/Gazebo Integration

The Publish/Subscribe Architecture

The interplay between Gazebo and the Robot Operating System (ROS) relies on a standardized, asynchronous publish/subscribe architecture. Simulated sensors attached to the AUV model generate data packages within Gazebo and publish them to specific ROS topics. Conversely, the AUV's autonomous control nodes subscribe to these topics to ingest state data, execute control algorithms, and publish calculated thrust vectors back to actuator topics.



The Transition to ROS 2

The marine robotics field is undergoing a systematic migration from ROS 1 to ROS 2, which utilizes a decentralized Data Distribution Service (DDS) communication layer. While older architectures like the original `uuv_simulator` relied heavily on the master-node communication of ROS 1, modern simulations utilize **Project Plankton**—the official open-source port of `uuv_simulator` built specifically for ROS 2 and the modernized Ignition Gazebo ecosystem. This transition introduces real-time processing guarantees, deterministic node execution, and optimized multi-robot network topologies, which are critical for simulating synchronized multi-AUV swarms.

2.5 Advantages and Limitations

Factor	Gazebo Simulation Advantages	Gazebo Simulation Limitations
Ecosystem	Compatible with ROS/ROS 2; extensive library of marine assets.	Dependency on middleware version matching; complex installation configurations.

Physics	Handles 6-DOF rigid-body kinetics and hydrodynamic overlays.	Hydrodynamic coefficients are idealized; struggles with transient fluid phenomena (e.g., turbulent wakes).
Validation	Zero-risk environment for training RL loops and testing SLAM algorithms.	"Sim-to-real" gap; perception models often fail under non-uniform real-world turbidity.
Efficiency	Headless deployment (gzserver) enables parallel computing.	High-fidelity ray-tracing for acoustic payloads creates computational bottlenecks.

3.AUV Dynamic Modelling in Gazebo

3.1 Coordinate systems and 6-DOF Model

To accurately track and control a submerged tactical asset, the navigation software must continuously translate data between two distinct mathematical reference frames. Understanding how these frames interact is the foundation of the vehicle's entire guidance, navigation, and control (GNC) stack.

3.1. Coordinate Systems and 6-DOF Model

Underwater mapping and localization rely heavily on the continuous transformation between a global map and the vehicle's localized perspective:

- **Inertial / Earth-Fixed Frame (X_E, Y_E, Z_E):** Think of this as the static, global map. The origin is usually pinned to a fixed geographic location, such as the initial deployment coordinates from a naval vessel or a harbor dock. Engineers use this stationary frame to define mission waypoints, plot the overall trajectory, and measure the vehicle's absolute positional drift over time.
- **Body-Fixed Frame (X_B, Y_B, Z_B):** This is a dynamic, moving coordinate system physically attached to the AUV itself, with its origin placed directly at the vehicle's center of gravity (CG). This frame is critical for the flight software because onboard sensors like IMUs and Doppler Velocity Logs (DVL) measure data relative to the vehicle's current orientation. Furthermore, all thruster forces and control surface deflections are computed and applied within this localized frame.

The transformation between these frames defines the vehicle's 6-Degree-of-Freedom (6-DOF) state, encompassing three translational movements (surge, sway, heave) and three rotational movements (roll, pitch, yaw).

3.2. Rigid-Body Equations of Motion

Gazebo simulation physics incorporates the non-linear Newton-Euler equations of motion. AUVs require full dynamic formulation to account for hydrodynamic coupling:

- **MRB (Rigid-Body Mass and Inertia Matrix):** Represents the physical mass and distribution of the vehicle.

$$M_{RB}\dot{\nu} + C_{RB}(\nu)\nu = \tau_{actuator} + \tau_{hydrodynamic}$$

Breaking this down into its practical engineering components:

- **M_{RB} (Rigid-Body Mass and Inertia Matrix):** This accounts for the physical dry weight of the vehicle and how its mass is distributed across the hull geometry.
- **$C_{RB}(\nu)$ (Coriolis and Centripetal Matrix):** As the AUV rotates—especially during aggressive tactical maneuvers like sharp yaw turns—this matrix calculates the fictitious forces that act on the vehicle due to the rapid rotation of the body-fixed frame.
- **ν and $\dot{\nu}$ (Velocity and Acceleration):** These vectors represent the vehicle's linear and angular speeds, as well as their rate of change over time.
- **$\tau_{actuator}$ (Control Inputs):** This represents the active mechanical forces and torques commanded by the autopilot, such as propeller thrust or the mechanical deflection of steering rudders.
- **$\tau_{hydrodynamic}$ (External Environment):** This captures how the ocean pushes back. It encompasses the drag, buoyancy, and added mass forces exerted by the surrounding water, which fundamentally alter how the vehicle handles and accelerates.

4. Hydrodynamic Modelling in Gazebo

4.1. Added Mass Matrix

Because water is a dense, incompressible fluid medium, an accelerating AUV must physically push away and displace a volume of water equal to its own directional motion. This fluid resistance creates an inertial lag known as "Added Mass." In Gazebo, this is implemented by introducing a non-diagonal 6x6 Added Mass matrix M_A that scales with the hull geometry. This is mathematically combined with the rigid-body mass matrix to dictate true acceleration properties:

$$M_{Total} = M_{RB} + M_A$$

4.2. Hydrodynamic Damping

Hydrodynamic damping models energy loss due to skin friction and vortex shedding. The Gazebo plugin calculates drag by combining linear damping matrices (dominant at low speeds) and quadratic drag tensors (dominant during high-speed maneuvers). These forces are computed dynamically based on the vehicle's velocity relative to environmental ocean currents.

4.3. Restoring Forces (Buoyancy and Gravity)

A defense-grade AUV cannot rely purely on software and thrusters to stay upright; it must be inherently stable in the water even if the power fails. This submerged stability is dictated by hydrostatic restoring forces, which rely on the continuous tug-of-war between two fundamental physical properties:

- Gravity ($W = mg$): The total physical weight of the vehicle, pulling directly downward through its Center of Gravity (CG).
- Buoyancy ($B = \rho g \nabla$): The upward lift exerted by the displaced water, acting through the vehicle's geometric Center of Buoyancy (CB) in accordance with Archimedes' principle. (*Note: ρ represents water density, and ∇ represents the displaced volume*).

For stability, naval engineers ensure the Center of Buoyancy is positioned vertically above the Center of Gravity. This structural offset generates a metacentric restoring torque, which automatically returns the vehicle to a level orientation following external disturbances, serving as a passive safety mechanism.

Because the lifting force (buoyancy) sits above the pulling force (gravity), the offset acts exactly like a heavy pendulum. Together, they generate a natural "metacentric restoring torque" that automatically swings the vehicle back to a stable, level orientation—without requiring a single line of code, thruster input, or battery power.

5. Literature Survey

5.1. The Evolution of Core Physics and Middleware Frameworks

Early autonomous underwater vehicle (AUV) research often relied on physical testbeds or highly simplified kinematic models that could not accurately capture the complexities of the marine environment (Brown et al., 2009) [4]. The introduction of advanced physics engines marked a major turning point in marine simulation. Manhães et al. (2016) [1] established a critical foundation by extending Gazebo's terrestrial physics engine into the underwater domain through the open-source `uuv_simulator` package. By developing custom C++ plugins, they enabled the precise calculation of 6-DOF rigid-body equations, hydrostatic effects, and hydrodynamic damping without being strictly tethered to a specific middleware.

While ROS remains the dominant middleware, researchers have successfully adapted Gazebo to other architectures to meet specific timing constraints. For example, Watanabe et al. (2015) [3] demonstrated how integrating Gazebo with the Robot Construction Kit (Rock) framework enables deterministic, real-time Hardware-In-the-Loop (HIL) simulations, a necessity for validating complex, time-sensitive hardware architectures. Today, the field is undergoing another architectural shift, moving from ROS 1 frameworks toward ROS 2 ecosystems (such as Project Plankton) and utilizing alternative hydrodynamic approaches like the `freefloating_gazebo` package to better model vehicles crossing the air-water interface.

5.2. Sensor Simulation and Environmental Perception

Because regular electromagnetic signals fail underwater, AUVs rely on specialized perception sensors that are notoriously difficult to simulate. Recent literature highlights a massive shift toward high-fidelity sensor modeling to bridge the "sim-to-real" gap. The DAVE (Aquatic Virtual Environment) project, detailed by Zhang et al. (2022) [5], builds upon the foundational physics of earlier simulators by integrating GPU-accelerated models for multibeam sonars and Doppler Velocity Loggers (DVL). Crucially, this framework allows researchers to import real-world, high-resolution bathymetry and simulate water turbidity, challenging computer vision algorithms with realistic environmental noise.

Simulating these noisy environments is essential for practical deployment. Sang and Norris (2023) [7] utilized AUV simulations to develop fuzzy sensor fusion techniques specifically for underwater pipeline inspection, demonstrating how virtual environments can safely test an AUV's ability to interpret degraded sensor data. However, native photorealistic visual rendering remains a persistent challenge; Watanabe et al. (2015) [3] noted that standard Gazebo graphics often fall short for advanced optical perception, requiring external libraries like `osgOcean` to accurately render light refraction, absorption, and scattering.

5.3. Intelligent Control and Multi-Agent (Swarm) Systems

As foundational physics and sensor simulations have matured, researchers are increasingly leveraging Gazebo to train computationally heavy artificial intelligence models. Advanced machine learning techniques require millions of trial-and-error iterations, which are impossible to execute safely in the physical ocean (Lakhyari et al., 2024) [2].

Q. Zhang et al. (2020) [6] demonstrated this capability by using deep interactive reinforcement learning (RL) to teach an AUV complex path-following behaviors in a simulated fluid environment. Beyond single-vehicle control, the focus of marine robotics is rapidly expanding to include multi-vehicle operations. Sampathkumar et al. (2022) [8] successfully utilized simulation environments to model multiple AUVs executing cooperative plume tracing. Modeling swarms introduces unique challenges, as the simulation must accurately restrict communication between agents to replicate the low bandwidth and high attenuation of real underwater acoustic modems.

5.4. Current Research Gaps Addressed by Modern Simulation

While current iterations of Gazebo excel at computing rigid-body dynamics and integrating seamlessly with ROS, the literature reveals distinct gaps that are actively being researched. The transition from idealized mathematical models to handling turbulent fluid phenomena—such as boundary-layer interactions and non-uniform turbidity—requires continuous improvement. Furthermore, as demonstrated by the diverse applications ranging from deep-sea mapping to offshore pipeline inspection, future simulation frameworks must balance the extreme computational load of ray-traced acoustic sensors with the need for high-speed, parallel execution required by deep learning and swarm development.

6. Conclusion

6.1. Summary of AUV Simulation Benefits

Developing and Deploying Autonomous Underwater Vehicles (AUVs) is a complex, risky, and expensive process. Simulation has proven to be an indispensable step in modern robotics projects, as it saves significant time and resources during the development phase by reducing hardware requirements and the need for frequent, expensive field test deployments. Through simulation frameworks like Gazebo, researchers can safely test algorithmic responses to harsh underwater environments without risking the loss of physical prototypes. Gazebo's modular architecture, specifically its ability to integrate with prominent robotics frameworks like ROS and Rock, provides a powerful tool for Hardware-in-the-Loop (HIL) simulations. By allowing developers to accurately model 6-DOF rigid-body equations, hydrodynamic damping, and complex sensor payloads like Doppler Velocity Logs (DVL) and multibeam sonars, Gazebo empowers engineers to validate guidance, navigation, and control systems before a vehicle ever touches the water.

6.2. Future Scope

While current simulation environments provide robust physics and kinematic testing, the trajectory of AUV research points toward several areas requiring advanced simulation capabilities:

- **More Realistic Visuals:** As AUVs rely increasingly on cameras for navigation, we need simulators that can generate photorealistic underwater environments out of the box. Future tools should natively handle tricky effects like water turbidity, light refraction, and scattering without depending on extra graphical plugins.
- **Better Coordination for Multi-Vehicle Systems:** As interest grows in using AUV swarms for wide-area surveys, our simulators need to better model the harsh reality of underwater communication—specifically low bandwidth and high signal attenuation. This is crucial for accurately testing how teams of robots share data and coordinate in real-world scenarios.
- **Support for Soft and Biomimetic Designs:** AUVs are beginning to take inspiration from marine life to improve maneuverability and energy efficiency. Future simulation frameworks must move beyond standard rigid-body mechanics to accurately test soft actuators and biomimetic fin propulsion.
- **Advanced AI and Sensor Fusion Training:** Because deep learning and sensor fusion are becoming essential for navigating complex, unstructured environments, simulators will serve as the primary training ground. Future systems must be optimized to handle the heavy computational load required to train these advanced AI models.

7.References

- [1] M. M. M. Manhães, S. A. Scherer, M. Voss, L. R. Douat, and T. Rauschenbach, "UUV Simulator: A Gazebo-based package for underwater intervention and multi-robot simulation," in *IEEE/MTS OCEANS*, Monterey, CA, USA, 2016, pp. 1-8.
- [2] A. Lakhyari, M. Boulaala, Z. Mekrini, and M. E. Mrabet, "Autonomous underwater vehicle: An overview of current technologies and applications," in *International Conference on Industrial Systems Engineering and Energy Conversion (ICISEEC'24)*, Tangier, Morocco, July 13, 2024.
- [3] T. Watanabe, G. Neves, R. Cerqueira, T. Trocoli, M. Reis, S. Joyeux, and J. Albiez, "The Rock-Gazebo integration and a real-time AUV simulation," in *Proceedings of the 12th Latin American Robotics Symposium (LARS) and 3rd Brazilian Symposium on Robotics (SBR)*, Salvador, Brazil, 2015, pp. 132-138.
- [4] H. C. Brown, A. Kim, and R. M. Eustice, "An overview of autonomous underwater vehicle research and testbed at PeRL," Perceptual Robotics Laboratory, University of Michigan, Tech. Rep., 2009.
- [5] C. Zhang, J. McKee, J. McCammon, J. Splinter, J. Vance, and G. Sharp, "DAVE aquatic virtual environment: Toward a general underwater robotics simulator," in *IEEE/MTS OCEANS*, Hampton Roads, VA, USA, 2022, pp. 1-10.
- [6] Q. Zhang, J. Lin, Q. Sha, and G. Li, "Deep interactive reinforcement learning for path following of autonomous underwater vehicle," *IEEE Access*, vol. 8, pp. 24258-24268, 2020.
- [7] I. Sang and W. Norris, "An autonomous underwater vehicle simulation with fuzzy sensor fusion for pipeline inspection," *IEEE Sensors Journal*, 2023.
- [8] A. Sampathkumar et al., "Plume tracing simulations using multiple autonomous underwater vehicles," in *Proceedings of the 16th International Conference on Underwater Networks & Systems (WUWNet)*, Boston, MA, USA, 2022, pp. 1-5.